Multidimensional Goodhart

Measurement Dimensions, Exchange Rates, and Hidden Harm — Technical Abstract

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Abstract. How do measurement dimensions affect hidden harm? Multidimensional Goodhart is not a theorem that more metrics help, hurt, or make residuals complex. In the one closed model where the question has a complete answer, per-agent fixed-deficit hidden harm is conserved under changes of the measured set exactly when every active gaming channel does the same hidden harm per score point (an additive exchange-rate iff-condition); otherwise harm is reduced, increased, or rerouted through the channels the design makes cheap, and the dimension count never enters by itself. The result is licensed by a response-modeling contract: proxy pressure supports a calculation only after the response channel, action/search geometry, aggregation rule, hidden value or harm model, and falsifier are declared. Supporting closed results — pure-selection δ-reweighting bounds, a quadratic Stackelberg wedge, a convex score-deficit budget, and a deterministic adaptive-hardening boundary — say when the exchange-rate frame applies and what feeds it. The main negative result is equally important: score movement alone does not identify mechanism or welfare.

The Measurement-Dimension Question

Should a gamed scorecard measure more things, fewer things, or different things — and will the hidden damage shrink, grow, or just move? Scalar Goodhart warnings perform two compressions and answer neither of the questions underneath them [1, 5, 15]. First, the target is reduced to a proxy. Second, the residual between proxy and target is treated as an unnamed loss. In multidimensional settings the residual is the object of interest: it has direction, support, exchange rates, active constraints, and evidence requirements. The question is therefore not whether proxy pressure can break a measure, but which response model makes a proposed calculation about hidden movement legal — and what that calculation says about the measured set.

Write $G : S \to \mathbb{R}^m$ for target-relevant state, $P : S \to \mathbb{R}^k$ for proxy features, φ for the intended proxy relation, and $\varepsilon(s) = P(s) - \varphi(G(s))$ for residual proxy artifact. A claim must also say how policy exposure changes the population or behavior: selection changes weights $W_\theta(u)$ over fixed types, while intervention changes the fixed-type response kernel $K_\theta(ds \mid u)$.

No Generic Law

- Dimension count alone does not determine hidden harm.
- “More measured dimensions” has no sign without an aggregation rule and harm exchange rates.
- Signed aggregate hidden error does not measure damage: positive and negative hidden movements can cancel before a value functional is named.
- Baseline covariance is only a local velocity, not a finite-pressure selection primitive.
- Absolute continuity is not the causal intervention boundary.
- The selection/intervention split is not representation-free or marginally identifiable.
- Coordinate-explicit selection bounds are not welfare or coordinate-free claims until a value metric is declared.
- Convex affordability is not a hidden-harm bound.
- Generic minimum-complexity attraction is not a theorem.

Contract

The response-modeling contract is one paragraph: declare the type space U, baseline law and behavior, policy exposure, selection weights and/or response kernels, action/search geometry and stakes when interventions are claimed, proxy/target relation, aggregation rule, hidden value or harm model, evidence standard, and falsifier. Without those primitives, a Goodhart explanation is a retrospective label. With them, the framework says which of the following calculations can be imported and what it refuses to infer.

The Exchange-Rate Diagnostic

With additive score over a measured channel set M, separable quadratic costs, fixed score deficit d, and declared hidden harm rates h_j, the per-agent hidden harm of the cost-minimizing response is

$$H_M(d) = d\frac{\sum_{j\in M} h_j \kappa_j w_j}{\sum_{j\in M} \kappa_j w_j^2},$$

a weighted harm-per-score average. Fixed-deficit per-agent harm is conserved across all measured sets drawn from a channel pool iff $h_j = cw_j$ on that pool; otherwise a measured-set change reduces, increases, or reroutes harm through the channels it makes cheap. Conjunctive aggregation, population entry, and undeclared harm rates each break the transfer.

Supporting Calculations

- **Selection bounds.** For pure selection with $\delta = \left\| d\frac{\mu_\theta}{d}\mu_0 - 1 \right\|_{L^2(\mu_0)}$ and drift $B_H = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu_0}[H]$, hidden coordinates satisfy $\left|B_{H_i}\right| \le \delta s_i$ and, after declaring the Euclidean coordinate norm, $\|B_H\|_2 \le \delta \|s\|_2$. With a declared scalar value vector v, $|v \cdot B_H| \le \delta\sqrt{v^T \Sigma_H v}$. The value metric is an input, not learned from score movement. Selection is the channel that must be excluded or bounded before any intervention reading.

- **Intervention budgets.** In the one-dimensional quadratic threshold toy, gaming is worthwhile exactly within the wedge $\Delta = \sqrt{2\kappa V}$. More generally, with convex cost c and linear proxy gain $w \cdot a$, the private score-deficit cost is $m(d) = \sup_{\lambda \geq 0}[\lambda d - c^*(\lambda w)]$. This is not a welfare bound without hidden harm; it supplies the "which channels are cheap" input the exchange-rate diagnostic prices.
- **Adaptive hardening.** Under fixed finite M, fixed d, fixed V, fixed weights, additive proxy gain, separable quadratic costs, deterministic observation, and monotone capacity hardening, gaming is feasible exactly when $S_t(M) = \sum_j \kappa_{j,t} w_j^2 \geq \frac{d^2}{2V}$ and stops exactly below that threshold.

Literature Relation

Goodhart, Campbell, Strathern, and Manheim–Garrabrant provide the genealogy and taxonomy [1, 5, 12, 15]. Lucas, strategic classification, performative prediction, multitask incentives, reward hacking, adaptive holdout work, and scalar tail-conditioned Goodhart provide formal analogues [3, 4, 6, 7, 10, 11, 13, 14], but each supplies only some primitives. On the empirical side, the hospital-readmissions and education-accountability literatures supply realized harm-per-score averages and score-side action traces, but not the harm-side designs that would identify channel-level exchange rates [2, 8, 9, 16, 17]. The framework's claim is reduction rather than unification: name the primitive a source contributes, and name what it omits.

Falsifiers, Refusals, and the Open Problem

The framework would be falsified by a domain where primitives are declared in advance, response channel and action traces distinguish nearby mechanisms, and the licensed calculation systematically predicts the wrong response shape or direction while a simpler score-only rule predicts it correctly. Narrow theorem failures would also falsify the relevant import: selection drift exceeding the δs_i envelope under its hypotheses; deterministic hardening violating the $S_t(M) < \frac{d^2}{2V}$ boundary; or additive fixed-deficit harm ignoring the $h_j = cw_j$ exchange-rate condition.

The framework refuses policy and welfare conclusions when primitives are unavailable. Examples include drifting type spaces, unobservable scorecard weights, changing measured sets, no defensible hidden value model, or applications where score movement is the only evidence.

The channel-identification question, by contrast, now has a toy-model answer: with harm linear in actions, the rate vector h is identifiable exactly when the action profiles induced across policy regimes have full column rank — regimes differing only in stringency are rank-deficient — so what remains open there is empirical, not mathematical.

The signature open problem is the residual-shape conjecture: under what declared response geometry, update rule, and predeclared complexity functional does repeated proxy repair drive hidden failure toward a predictable residual shape? The current answer is negative without those declarations.

References

[1] Donald T. Campbell. 1979. Assessing the impact of planned social change. *Evaluation and Program Planning* 2, 1 (1979), 67–90.

[2] Kumar Dharmarajan, Yongfei Wang, Zhenqiu Lin, Sharon-Lise T. Normand, Joseph S. Ross, Leora I. Horwitz, Nihar R. Desai, Lisa G. Suter, Elizabeth E. Drye, Susannah M. Bernheim, and Harlan M. Krumholz. 2017. Association of Changing Hospital Readmission Rates With Mortality Rates After Hospital Discharge. *JAMA* 318, 3 (2017), 270–278. https://doi.org/10.1001/jama.2017.8444

[3] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. 2015. Generalization in Adaptive Data Analysis and Holdout Reuse. In *Advances in Neural Information Processing Systems*, 2015. Retrieved from https://arxiv.org/abs/1506.02629

[4] El-Mahdi El-Mhamdi and Lê-Nguyên Hoang. 2024. On Goodhart's Law, with an Application to Value Alignment. Retrieved from https://arxiv.org/abs/2410.09638

[5] Charles A. E. Goodhart. 1975. Problems of Monetary Management: The U.K. Experience. *Papers in Monetary Economics, Volume I, Reserve Bank of Australia.*

[6] Moritz Hardt, Nimrod Megiddo, Christos Papadimitriou, and Mary Wootters. 2016. Strategic Classification. In *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science (ITCS)*, 2016. 111–122.

[7] Bengt Holmström and Paul Milgrom. 1991. Multitask Principal-Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design. *The Journal of Law, Economics, and Organization* 7, Special Issue (1991), 24–52. https://doi.org/10.1093/jleo/7.special_issue.24

[8] Brian A. Jacob. 2005. Accountability, Incentives and Behavior: The Impact of High-Stakes Testing in the Chicago Public Schools. *Journal of Public Economics* 89, 5–6 (2005), 761–796. https://doi.org/10.1016/j.jpubeco.2004.08.004

[9] Brian A. Jacob and Steven D. Levitt. 2003. Rotten Apples: An Investigation of the Prevalence and Predictors of Teacher Cheating. *The Quarterly Journal of Economics* 118, 3 (2003), 843–877. https://doi.org/10.1162/00335530360698441

[10] Robert E. Lucas. 1976. Econometric Policy Evaluation: A Critique. *Carnegie-Rochester Conference Series on Public Policy* 1, (1976), 19–46. https://doi.org/10.1016/S0167-2231(76)80003-6

[11] Adrien Majka and El-Mahdi El-Mhamdi. 2025. The Strong, Weak and Benign Goodhart's Law. An Independence-Free and Paradigm-Agnostic Formalisation. Retrieved from https://arxiv.org/abs/2505.23445

[12] David Manheim and Scott Garrabrant. 2018. Categorizing Variants of Goodhart's Law. *arXiv preprint arXiv:1803.04585* (2018).

[13] Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dünner, and Moritz Hardt. 2020. Performative Prediction. In *Proceedings of the 37th International Conference on Machine Learning (Proceedings of Machine Learning Research)*, 2020. PMLR, 7599–7609. Retrieved from https://proceedings.mlr.press/v119/perdomo20a.html

[14] Joar Skalse, Nikolaus H. R. Howe, Dmitrii Krasheninnikov, and David Krueger. 2022. Defining and Characterizing Reward Gaming. In *Advances in Neural Information Processing Systems*, 2022.

[15] Marilyn Strathern. 1997. `Improving ratings': audit in the British University system. *European Review* 5, 3 (1997), 305–321.

[16] Rishi K. Wadhera, Karen E. Joynt Maddox, Jason H. Wasfy, Sebastien Haneuse, Changyu Shen, and Robert W. Yeh. 2018. Association of the Hospital Readmissions Reduction Program With Mortality Among Medicare Beneficiaries Hospitalized for Heart Failure, Acute Myocardial Infarction, and Pneumonia. *JAMA* 320, 24 (2018), 2542–2552. https://doi.org/10.1001/jama.2018.19232

[17] Rachael B. Zuckerman, Steven H. Sheingold, E. John Orav, Joel Ruhter, and Arnold M. Epstein. 2016. Readmissions, Observation, and the Hospital Readmissions Reduction Program. *New England Journal of Medicine* 374, 16 (2016), 1543–1551. https://doi.org/10.1056/NEJMsa1513024